

Gold-Standard Annotation Guidelines — Pashto Stemmer

Goal: produce trustworthy ground-truth **stems** for the most frequent Pashto words so we can report real accuracy and Paice over/under-stemming indices.

The file `gold_annotation_template.csv` is **pre-filled** with the stemmer's best guess. You only need to correct what is wrong — minimal typing.

Columns

Column	Meaning	What to do
<code>word</code>	the normalized surface word	do not change
<code>frequency</code>	corpus frequency	do not change
<code>proposed_stem</code>	the stemmer's output	do not change (reference)
<code>correct_stem</code>	the true light stem	edit ONLY if <code>proposed_stem</code> is wrong
<code>pos</code>	guessed part of speech (N/ADJ/V)	fix if clearly wrong
<code>is_irregular</code>	1 if a suppletive/irregular verb form	set 1/0
<code>is_compound</code>	1 if a closed compound	set 1/0
<code>confidence</code>	stemmer confidence	review low ones first
<code>notes</code>	free text	optional

What is a "light stem" here?

Reduce a word to its **most basic form that is still a real Pashto word**, removing inflection (number, gender, case, verb agreement) and clearly productive derivation. Do **not** reduce to an abstract trilateral root.

Examples of the target:

- کورونه (houses) → کور
- خبرونه (news) → خبر
- ښکلې / ښکلې (beautiful, m/f) → ښکلې
- غوښتل / غواړي (wants / wanted) → غوښتل (infinitive lemma for verbs)
- افغانستان (Afghanistan) → افغانستان (proper nouns stay whole)
- او، په، دا (function words) → unchanged (stopwords)

Conventions

- Verbs** are lemmatized to the **infinitive** (ends in ڼل): کول → کوي.
- Proper nouns** (names of people, countries, cities, organizations) are left **unchanged**.
- Stopwords / function words** are left **unchanged**.
- Keep the distinct feminine yeh letters (ې ۍ); write ې as ۍ (already normalized in the file).

5. If a word is ambiguous, choose the reading that fits general news text and add a note.

After annotating

Save the file (UTF-8) and run:

```
python experiments/evaluate_gold.py dataset/gold_annotation_template.csv
```

This prints exact-match **accuracy** and **Paice UI/OI/SW** for our stemmer and the baseline, using your **correct_stem** column as ground truth.